

ECE 222 Homework #3

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Calculate the electric potential for problems **4.19, 4.25, 4.26, 4.27, 4.28, 4.29** from the textbook.

Do problems **4.33, 4.34, 4.35, 4.36, 4.37, 4.38** from the textbook.

Regarding reference points for the first group of problems: A good reference point is one where the potential isn't blowing up.

For example, infinity is usually good for spherical potentials, but not for cylindrical (because of the $\ln$ dependence). Depending on the charge distribution, zero might be a good point, or some other characteristic position (e.g., radius of sphere or cylinder).

The shape of the curve will not change with the change of the reference point, it's just that the whole curve moves up or down when you change the reference point.

Here are suggested reference points for the first group of problems:

4.19: Pick coordinate origin as reference point

4.25: Pick $R = 0$, a , or infinity

4.26: Electric field doesn't integrate analytically, you have to integrate numerically (use programming language of your choice or a programmable calculator). Reference point at zero

4.27: Pick r_1 or r_2 as reference r (everything between 0 and r_1 has the same potential as field is zero in that range)

4.28: Pick $R = 0$ as reference point

4.29: Either region from 0 to a (will all have the same potential) or infinity

*4.19

E_T only acts on P_0 in the $\hat{y}$ direction

Three infinite lines of charge, all parallel to the z axis, are located at the three corners of the kite-shaped arrangement shown in Fig. P4.19. If the two right triangles are symmetrical and of equal corresponding sides, show that the electric field is zero at the origin. By symmetry, $E_{s_{ox}}$ & $E_{s_{ox}}$ cancel

Figure P4.19 Kite-shaped arrangement of line charges for Problem 4.19.

$$\therefore \cos \theta = \frac{y}{d}$$

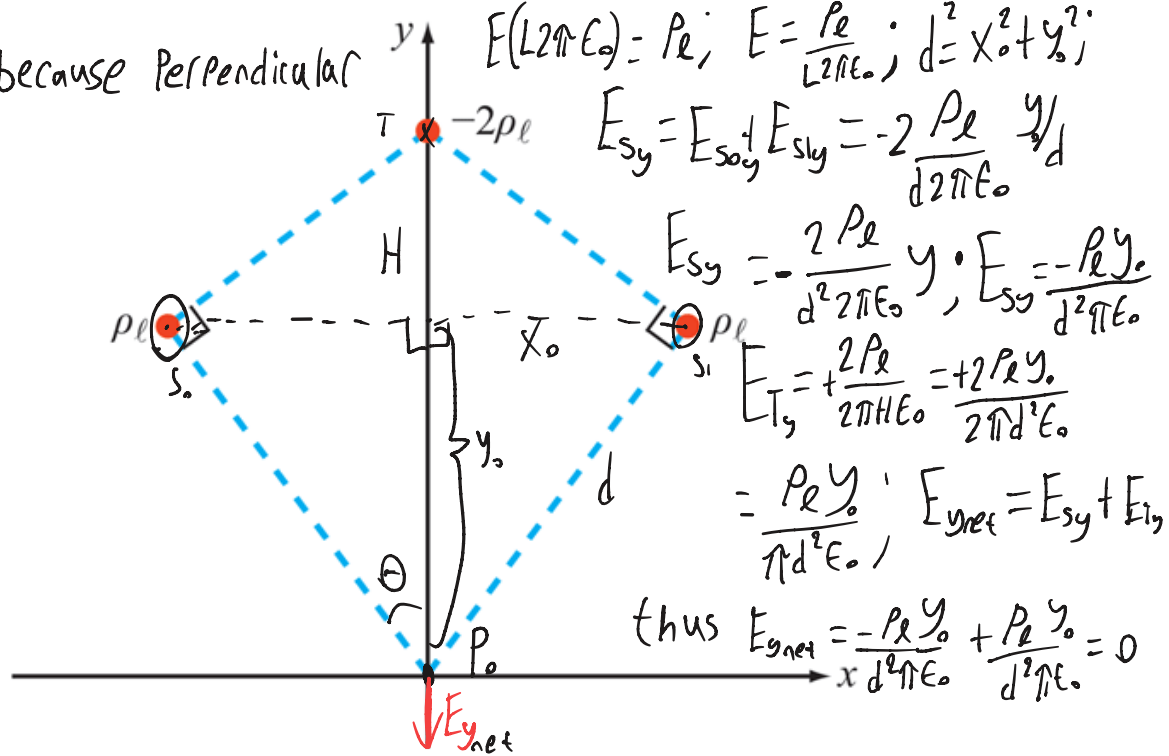
$$\vec{S}_0 T \cdot \vec{S}_0 O = 0 \text{ because perpendicular}$$

$$[-x_0, -y_0] \cdot [x_0, H-y_0]$$

$$= x_0^2 + y_0^2 - H y_0 = 0;$$

$$x_0^2 + y_0^2 = H y_0;$$

$$\therefore H = \frac{x_0^2 + y_0^2}{y_0} = \frac{d^2}{y_0}$$



$$V_{origin} = 0$$

$$E = \frac{D}{\epsilon}; \quad V(B) - V(A) = -\int_A^B E \cdot d\ell$$

- *4.25 The electric flux density inside a dielectric sphere of radius a centered at the origin is given by

see pg. 4 $V(a)=0; V(R)-V(a) = -\int_a^R \frac{\rho_0 R}{\epsilon} dR$

$$\mathbf{D} = \hat{\mathbf{R}}\rho_0 R \quad (\text{C/m}^2) = \frac{\rho_0}{2\epsilon}(a^2 - R^2) \quad R \leq a$$

where ρ_0 is a constant. Find the total charge inside the sphere.

- 4.26 In a certain region of space, the charge density is given in cylindrical coordinates by the function

$$\rho_v = 5re^{-r} \quad (\text{C/m}^3).$$

Apply Gauss's law to find $\mathbf{D}$.

- *4.27 An infinitely long cylindrical shell extending between $r = 1$ m and $r = 3$ m contains a uniform charge density ρ_{v0} . Apply Gauss's law to find $\mathbf{D}$ in all regions.

- 4.28 If the charge density increases linearly with distance from the origin such that $\rho_v = 0$ at the origin and $\rho_v = 4 \text{ C/m}^3$ at $R = 2$ m, find the corresponding variation of $\mathbf{D}$.

$$4.26. V(0)=0; V(r) = -\frac{5}{\epsilon} \int_0^r \frac{2 - (r^2 + 2(r+2)e^{-r})}{r} dr$$

$$4.27: V(r) = \begin{cases} 0 & r < 1 \\ -\frac{\rho_{v0}}{2\epsilon} \left[\frac{r^2-1}{2} - \ln r \right] & 1 \leq r \leq 3 \\ -\frac{\rho_{v0}}{2\epsilon} (4 - \ln 3) - \frac{4\rho_{v0}}{\epsilon} \ln(r/3) & r > 3 \end{cases}$$

$$4.28. V(R) = -\int_0^R \frac{R^2}{2\epsilon} dR; V(0)=0; V(R) = -\frac{R^3}{6\epsilon}$$

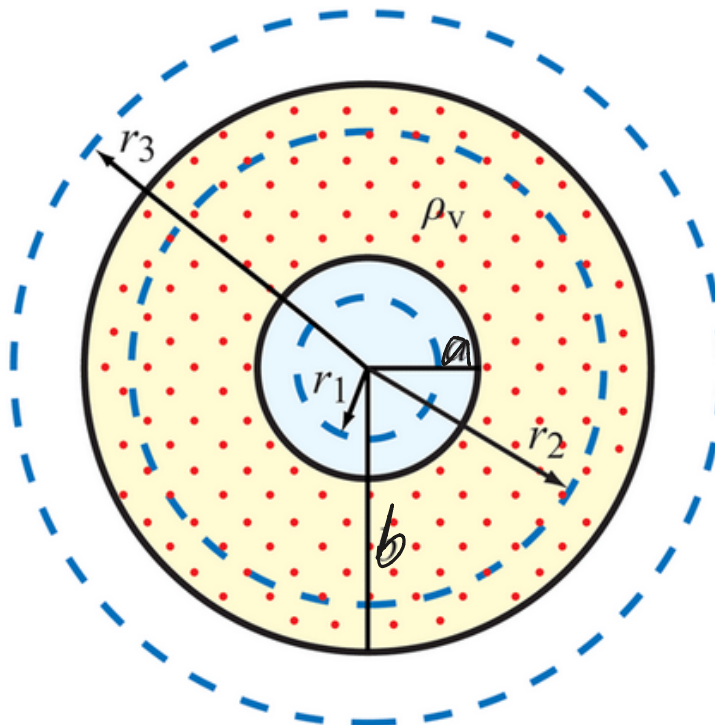
See Pg. 10

- 4.29 A spherical shell with outer radius b surrounds a charge-free cavity of radius $a < b$ (Fig. P4.29). If the shell contains a charge density given by

$$\rho_v = -\frac{\rho_{v0}}{R^2}, \quad a \leq R \leq b,$$

where ρ_{v0} is a positive constant, determine $\mathbf{D}$ in all regions.

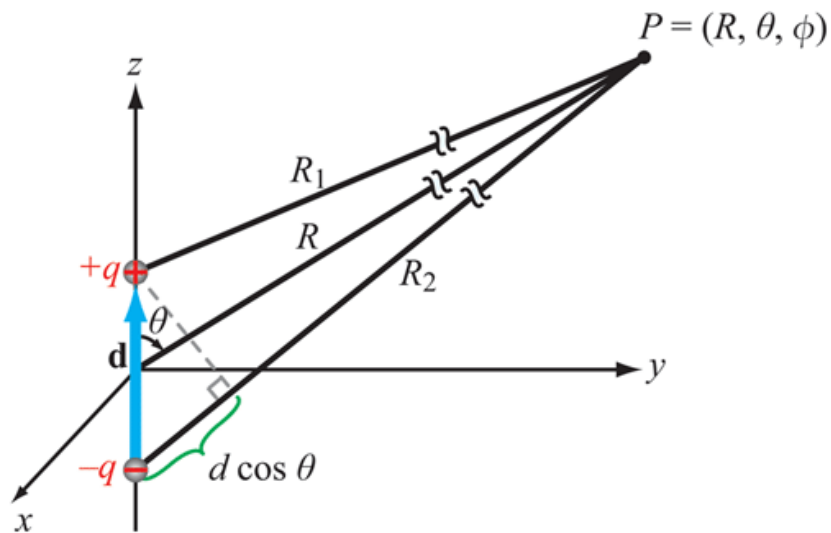
Figure P4.29 Problem 4.29.



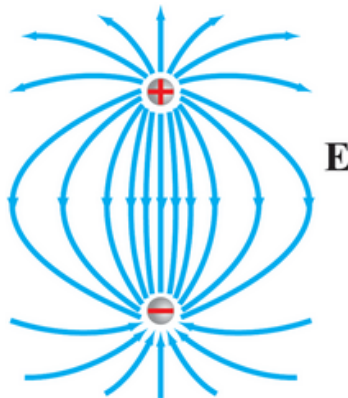
$$V(R) = \begin{cases} \rho_{v0}/\epsilon \ln \frac{a}{b}, & R < a \\ \frac{\rho_{v0}}{\epsilon} \left[\ln \frac{R}{b} + \frac{a}{R} - 1 \right], & a \leq R \leq b \\ -\frac{\rho_{v0}(b-a)}{\epsilon R}, & R > b \end{cases}$$

- 4.33 Show that the electric potential difference V_{12} between two points in air at radial distances r_1 and r_2 from an infinite line of charge with density ρ_ℓ along the z axis is $V_{12} = (\rho_\ell / 2\pi\epsilon_0) \ln(r_2/r_1)$.
- *4.34 Find the electric potential V at a location a distance b from the origin in the x - y plane due to a line charge with charge density ρ_ℓ and of length l . The line charge is coincident with the z axis and extends from $z = -l/2$ to $z = l/2$.
- 4.35 For the electric dipole shown in Fig. 4-15, $d = 1$ cm and $|\mathbf{E}| = 4$ (mV/m) at $R = 1$ m and $\theta = 0^\circ$. Find $\mathbf{E}$ at $R = 2$ m and $\theta = 90^\circ$. *See pg. 11*

Figure 4-15 Electric dipole with dipole moment $\mathbf{p} = q\mathbf{d}$ (Example 4-9).



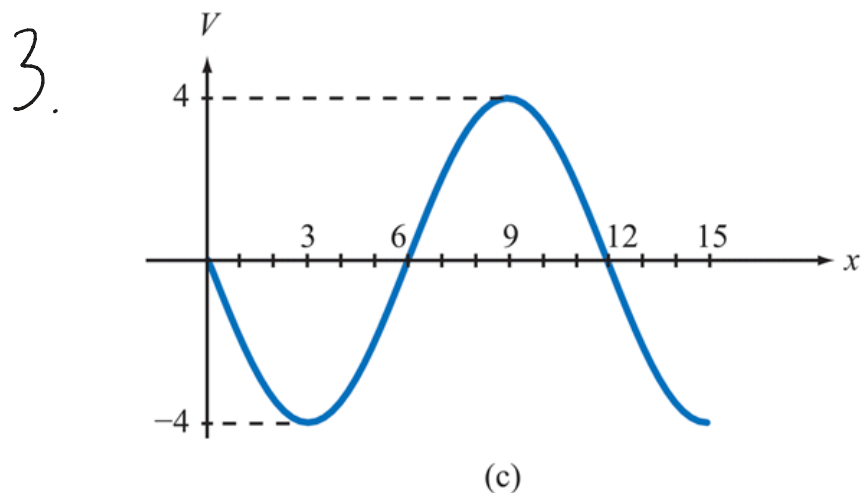
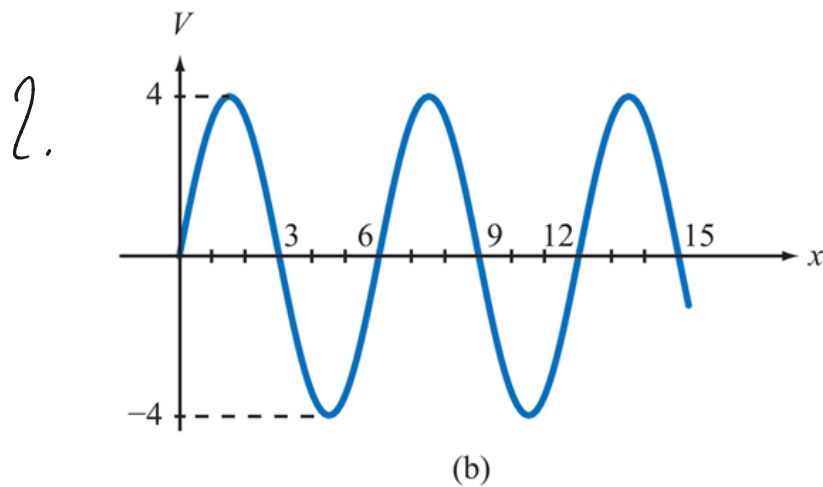
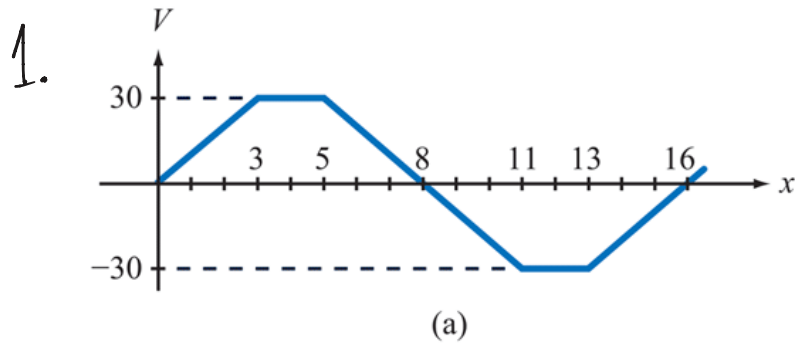
(a) Electric dipole



(b) Electric-field pattern

- 4.36 For each of the distributions of the electric potential V shown in Fig. P4.36, sketch the corresponding distribution of $\mathbf{E}$ (in all cases, the vertical axis is in volts and the horizontal axis is in meters).

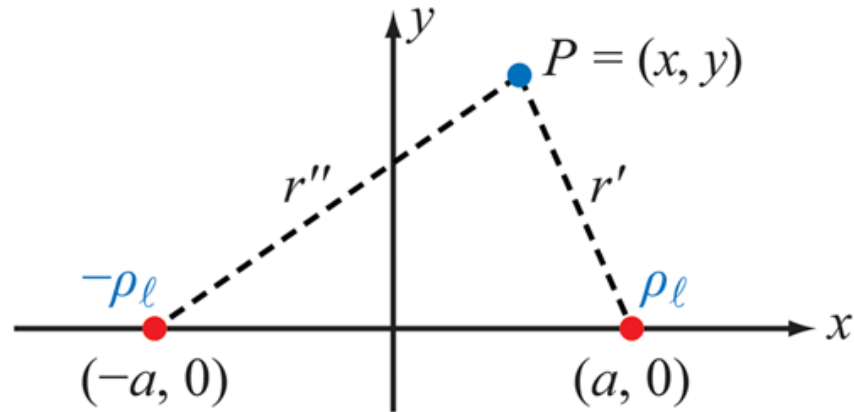
Figure P4.36 Electric potential distributions of Problem 4.36.



See Pg. 12

- *4.37 As shown in Fig. P4.37, two infinite lines of charge, both parallel to the z axis, lie in the x - z plane: one with density ρ_ℓ and located at $x = a$ and the other with density $-\rho_\ell$ and located at $x = -a$. Obtain an expression for the electric potential $V(x, y)$ at a point $P = (x, y)$ relative to the potential at the origin.

Figure P4.37 Problem 4.37.



- 4.38 Given the electric field

$$\mathbf{E} = \hat{\mathbf{R}} \frac{18}{R^2} \quad (\text{V/m})$$

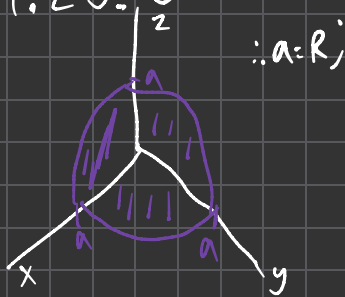
find the electric potential of point A with respect to point B where A is at $+2 \text{ m}$ and B at -4 m (both on the z axis).

Solutions to starred problems:

$$4.34 \blacksquare \quad V(b) = \frac{\rho_l}{4\pi\epsilon} \ln \left[\frac{l + \sqrt{l^2 + 4b^2}}{-l + \sqrt{l^2 + 4b^2}} \right] \text{ (V)}$$

$$4.37 \blacksquare \quad V = \frac{\rho_l}{2\pi\epsilon_0} \left[\ln \left(\frac{a}{\sqrt{(x-a)^2 + y^2}} \right) - \ln \left(\frac{a}{\sqrt{(x+a)^2 + y^2}} \right) \right]$$

4.25: $D = \hat{R} \rho_0 R = (C/m^2)$; Gauss $\rightarrow \oint \vec{D} \cdot d\vec{s} = Q_{enc}$; $\therefore SA_{sphere} = 4\pi a^2$; $Q_{enc} = \int \rho_0 a ds$



$\oint_S dS = 4\pi a^2$ $Q_{enc} = 4\pi \rho_0 a^3$

4.26: $\rho_v = 5r e^{-r} (C/m^3)$; Gauss $\rightarrow \oint_S \vec{D} \cdot d\vec{s}$; $\therefore dz \cdot dS = 0$; $\oint dS = 2\pi(rz)$ $Q_{enc} = D \cdot 2\pi(rz)$

$D = \frac{Q_{enc}}{2\pi(rz)}$; $Q_{enc} = \iiint_V \rho_v dV$ $dV = (r dr d\phi dz)$; $\int_0^{2\pi} \int_0^{r_0} \int_0^H 5r^2 e^{-r} dr d\phi dz$; $Q_{enc} = 10\pi H \int_0^{r_0} r^2 e^{-r} dr$

$U = r^2$ $dU = 2r dr$ $\therefore \int U dV = UV - \int V dU$

$dV = e^{-r} dr$ $V = -e^{-r}$; $\int r^2 e^{-r} = -r^2 e^{-r} - 2 \int r e^{-r} dr$ $U = r^2$ $dU = 2r dr$ $V = -e^{-r}$; $\int r^2 e^{-r} = r^2 e^{-r} + 2(-r e^{-r} - e^{-r}) = r^2 e^{-r} - 2(r e^{-r} + e^{-r})$

$Q_{enc} = 10\pi H \left[-r^2 e^{-r} - 2(r e^{-r} + e^{-r}) \right]_0^{r_0}$

$\int r e^{-r} = -r e^{-r} - \int -e^{-r} dr = -r e^{-r} - e^{-r}$

$= 10\pi H (2 e^{-r} (r^2 + 2r + 1))$; $D = \frac{5}{r} (r^3 + 2r + 1)(-e^{-r} + 2) \hat{r} C/m^2$

4.27: ρ_{v0} ; $1m \leq r \leq 3m$; for $r < 1 \rightarrow Q_{enc} = 0 = D$; for $1m \leq r \leq 3m$; $V_{cyl} = L\pi(r^2)$; $V_{shell} = L\pi(r^2 - 1)$

$Q_{enc} = V_{shell} \cdot \rho_{v0} = \rho_{v0} \pi(r^2 - 1)L$; $D_r(2\pi r L) = \rho_{v0} \pi(r^2 - 1)L$; $D(r) = \frac{\rho_{v0} \pi(r^2 - 1)}{2r} \hat{r}$; for $r > 3m \rightarrow Q_{enc} = \rho_{v0} \pi L(9 - 1) = 8\pi L \rho_{v0}$

$D_r(r)(2\pi r L) = 8\pi L \rho_{v0}$; $D_{r>3}(r) = \frac{4L \rho_{v0}}{r} \hat{r}$

$D = \begin{cases} 0 & r < 1m \\ \frac{\rho_{v0} \pi(r^2 - 1)}{2r} \hat{r} & 1m \leq r \leq 3m \\ \frac{4L \rho_{v0}}{r} \hat{r} & r > 3m \end{cases}$

4.28. $\rho_v = 0$ @ origin $\xrightarrow{\text{linearly}} \rho_v = 4 C/m^3$ @ $R = 2m$; $\therefore \rho_v = 2R C/m^3$; $Q_{enc} = \int_S D \cdot dS$; $Q_{enc} = D(4\pi R^2)$

$\frac{1}{r^2} \frac{d}{dr} (R^2 D_r) = \rho_v = 2R$; $\frac{d}{dr} (R^2 D_r) = 2R^3$; $R^2 D_r = \frac{R^4}{2} + C$; $\therefore C = 0$ for $\rho_v = 0$ @ $r = 0$

$D_r = R^2/2$

4.29: $\therefore D = Q_{enc} = 0$ for $a \leq R \leq b$; $D(4\pi R^2) = Q_{enc}$; $Q_{enc} = \int_V \rho_v dV$; $\nabla \cdot D = \frac{1}{R^2} \frac{d}{dR} (R^2 D_R) = \rho_v$
 integral form $\rightarrow dV = R^2 \sin \theta dR d\theta d\phi \rightarrow Q_{enc} = \int_0^{2\pi} \int_0^\pi \int_a^R -\rho_v \sin \theta dR d\theta d\phi = -2\pi \rho_v \int_0^\pi \sin \theta d\theta (R - a)$
 $-(R-a) 2\pi \rho_v \int_0^\pi \sin \theta d\theta = -4\pi (R-a) \rho_v$; $D(4\pi R^2) = -(R-a) \rho_v 4\pi$; $D = \frac{\rho_v (a-R)}{R^2}$

for $R > b \rightarrow Q_{enc} = -(b-a) 4\pi \rho_v \rightarrow D = \frac{\rho_v (a-b)}{R^2}$

$$D = \begin{cases} 0; & R < a \\ \rho_v (R-a)/R^2; & a \leq R \leq b \\ \rho_v (b-a)/R^2; & R > b \end{cases}$$

4.33: $E_{line} (2\pi r L) = \frac{Q_{enc}}{\epsilon}$; $\vec{E} = \frac{Q_{enc}}{2\pi r L \epsilon}$; $V_{12} = V_1 - V_2 = -\int_{r_2}^{r_1} \vec{E} \cdot d\vec{r}$; $\therefore dL = dr$; $V_{12} = \frac{Q_{enc}}{2\pi L \epsilon} \left[\ln(r) \right]_{r_2}^{r_1}$

$V_a = \frac{-Q_{enc}}{2\pi L \epsilon} [\ln r_1 - \ln r_2]$; $\therefore \frac{Q_{enc}}{L} = P_L$; $V_{12} = \frac{-P_L}{2\pi \epsilon} (\ln r_1 / r_2) = \frac{P_L}{2\pi \epsilon_0} (\ln r_2 / r_1)$

4.34: $V = \frac{1}{4\pi \epsilon_0} \int_L \frac{\rho_L dl}{R}$; $P = (b, 0, 0) \rightarrow R = \sqrt{b^2 + z^2}$; $\therefore dl = dz$; $dQ = P_L dz$; $V(b) = \frac{P_L}{4\pi \epsilon_0} \int_{-L/2}^{L/2} \frac{dz}{\sqrt{b^2 + z^2}}$

$\therefore \int \frac{dz}{\sqrt{b^2 + z^2}} = \ln(2 + \sqrt{z^2 + b^2})$ thus $V(b) = \frac{P_L}{4\pi \epsilon_0} \left[\ln(2 + \sqrt{z^2 + b^2}) \right]_{-L/2}^{L/2}$; $\frac{dz}{\sqrt{b^2 + z^2}}$

$V(b) = \frac{P_L}{4\pi \epsilon_0} \left[\ln\left(\frac{1}{2} + \sqrt{\frac{L^2}{4} + b^2}\right) - \ln\left(-\frac{1}{2} + \sqrt{\frac{L^2}{4} + b^2}\right) \right]$

$V(b) = \frac{P_L}{4\pi \epsilon_0} \ln \left[\frac{\sqrt{L^2/4 + b^2} + 1/2}{\sqrt{L^2/4 + b^2} - 1/2} \right]$

4.35: $d = 1 \text{ cm}$; $|E| = 4 \frac{\text{mV}}{\text{m}}$; $@ R = 1 \text{ m}$ & $\theta = 0^\circ$; $E(2 \text{ m}, 90^\circ) = ?$

$$E = \frac{\rho}{4\pi\epsilon_0 R^3} (2\cos\theta \hat{R} + \sin\theta \hat{\theta}); \quad \rho = qd$$

$$R_1 = 1, \theta_1 = 0 \rightarrow E(R_1, \theta_1) = E_1 = \frac{\rho}{4\pi\epsilon_0 (1)^3} (2\hat{R}); \quad \frac{4 \text{ mV}}{\text{m}} = \frac{2\rho}{4\pi\epsilon_0}$$

$$\frac{\rho}{4\pi\epsilon_0} = 2 \frac{\text{mV}}{\text{m}}; \quad E(R = 2 \text{ m}, \theta = 90^\circ) = E_2 = \frac{\rho}{4\pi\epsilon_0 (2)^3} (\hat{\theta})$$

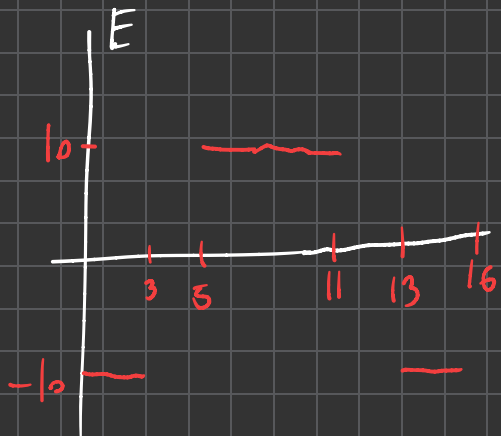
$$E_2 = (2 \frac{\text{mV}}{\text{m}}) \left(\frac{1}{8} \right) (\hat{\theta}) = \underline{250 \frac{\mu\text{V}}{\text{m}} \hat{\theta}}$$

4.36: $E = -\nabla V$; $E_x = -\frac{dV}{dx}$; 1. $-\frac{dV}{dx} = -10$ for $x \in [1, 3] \cup [13, 16]$; $\frac{dV}{dx} = 0$ for $x \in [3, 5] \cup [\frac{11}{3}, \frac{13}{3}]$

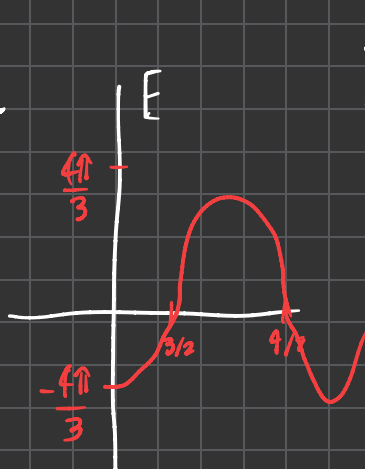
1. $E = -\frac{dV}{dx} = 10$ for $x \in [5, 11]$; 2. $V(x) = 4\sin(\frac{\pi x}{3})$; $E = -\frac{dV}{dx} = -\frac{4\pi}{3} \cos(\frac{\pi x}{3}) \text{ V/m}$

3. $V(x) = -4\sin(\frac{\pi x}{6})$; $\frac{dV}{dx} = \frac{2\pi}{3} \cos(\frac{\pi x}{6})$

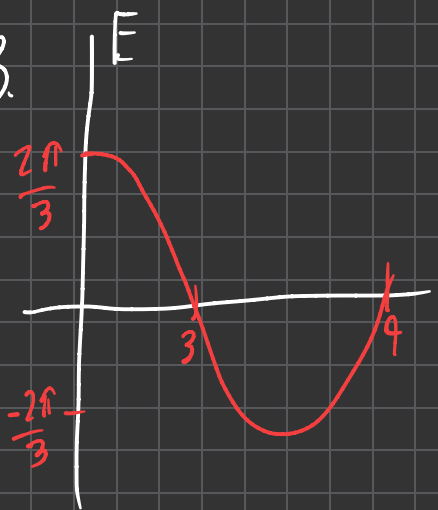
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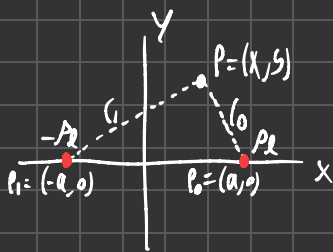
2.



3.



4.37.



$$|r_0| = \sqrt{(x-a)^2 + y^2}; |r_1| = \sqrt{(x+a)^2 + y^2}; E_t = E_0 + E_1;$$

$$E_0 = \frac{Q_0}{2\pi\epsilon_0 |r_0|^2}; E_1 = \frac{-Q_0}{2\pi\epsilon_0 |r_1|^2}; \therefore \hat{r}_0 = \frac{\langle x-a, y \rangle}{|r_0|}; \hat{r}_1 = \frac{\langle x+a, y \rangle}{|r_1|}$$

$$E_t = \frac{Q_0(x-a)}{2\pi\epsilon_0(x-a)^2 + y^2} \hat{x} + \frac{Q_0 y}{2\pi\epsilon_0(x-a)^2 + y^2} \hat{y} \\ + \frac{-Q_0(x+a)}{2\pi\epsilon_0(x+a)^2 + y^2} \hat{x} + \frac{-Q_0 y}{2\pi\epsilon_0(x+a)^2 + y^2} \hat{y}$$

$$V_{r_0}(P) = -\frac{Q_0}{2\pi\epsilon_0} \ln \frac{r_0}{a}; V_{r_1}(P) = \frac{Q_0}{2\pi\epsilon_0} \ln \frac{r_1}{a}; V_t(P) = V_{r_0} + V_{r_1}$$

$$= \frac{Q_0}{2\pi\epsilon_0} \left(\ln \frac{r_1}{a} - \ln \frac{r_0}{a} \right) = \frac{Q_0}{2\pi\epsilon_0} \left(\ln \frac{r_1}{r_0} \right) = \frac{Q_0}{2\pi\epsilon_0} \ln \left(\left(\frac{(x+a)^2 + y^2}{(x-a)^2 + y^2} \right)^{1/2} \right) \checkmark$$

$$4.38: E = \hat{r} \frac{18}{R^2}; V_{AB} = V_A - V_B = -\int_B^A E \cdot d\mathbf{l}; \therefore d\mathbf{l} = dR \hat{r}; V_{AB} = -\int_{-4}^A \frac{18}{R^2} \hat{r} dR$$

$$V_{AB} = -18 \hat{r} \left[-\frac{1}{R} \right]_{-4}^2 = -18 \left[-\frac{1}{2} - \left(-\frac{1}{-4} \right) \right] \hat{r} = -18 \left[-\frac{1}{2} - \frac{1}{4} \right] = \frac{18}{4} \hat{r} = 4.5V \hat{r}$$